PART 06 · CHAPTER 28

Case studies: follow the decision through

Successful trades, failed trades, and the decision to stand aside.

Learner edition: attempt first; continuation reference starts on a new page where supplied. Worked solutions and quiz answers are omitted.

advanced About 18 min of prose · practice takes additional time AI editorial review 2026-10-12 · v0.4

▼ Review scope

Full chapter final review with inherited initial reviewer/source evidence; final arithmetic, actual figures and native reader checks.

Recorded by /root. The review covers editorial evidence; it does not establish learning by human readers.

> THE IDEA TO KEEP
>
> A case is a chain from known data through a frozen decision to a later record. Keep favorable, unfavorable, skipped and unresolved cases together; a selected gallery cannot estimate performance.

By the end of this chapter

- Reconstruct a case from the information available at each cutoff.
- Process successful, failed, gapped, skipped and ambiguous opportunities under frozen rules.
- Apply the same accounting discipline to a range recovery and to timeframes that describe different intervals.
- Separate documented historical observations from fictional execution.

Builds on: Risk & position sizing · From chart idea to trade plan · A journal that can teach you something · Evidence before conviction · Statistics for a skeptical chart reader · Designing a complete trading system

First pass: four historical cutoffs

Attempt each cutoff before revealing the casebook, outcome headings or complete calendar ledger. These four windows were selected as teaching representatives after inspecting the fixed calendar set; they are not a blind sample or a performance trial. Selection details and all 36 windows remain in the continuation reference.

For each six-row window, compute the mean of the five context closes. Open an observation window only when the cutoff close is strictly above that mean. If it passes, freeze boundaries at 0.99 × cutoff close and 1.01 × cutoff close without rounding. At each of the next five archive bars, first compare the open with both boundaries; if strictly inside, compare the low/high. A sole upper touch is favorable, a sole lower touch adverse, both touches ambiguous; the first resolving observation ends the check. No touch by the fifth bar is unresolved. If the gate fails, keep the skip without a return or R. These are path labels, not trades.

At the cutoff, record eligibility, exact thresholds (if any), information available and what later data would be needed. Do not predict which path will occur. After the reveal, reconcile your original note with the archived continuation without editing the note.

Data are the preserved AAPL/MSFT/SPY educational archive, archive-adjusted USD with undocumented adjustment algorithm and exact session/time-zone. Depositor metadata declares CC0; upstream provenance and point-in-time vintage are limited. These rows are not certified contemporaneous quotes or executable fills. Full files, hashes and provenance are in data/historical/ and the continuation reference.

AAPL-2016-01 — context and cutoff only (archive-adjusted USD)

Archive date / clock	Open	High	Low	Close
2015-12-24 · context	104.93	104.93	103.93	104.01
2015-12-28 · context	103.59	103.69	102.22	102.83
2015-12-29 · context	102.97	105.35	102.87	104.68
2015-12-30 · context	104.53	104.64	103.18	103.32

Archive date / clock	Open	High	Low	Close
2015-12-31 · context	103.02	103.04	100.91	101.35
2016-01-04 · CUTOFF	98.786	101.46	98.211	101.44

MSFT-2016-02 — context and cutoff only (archive-adjusted USD)

Archive date / clock	Open	High	Low	Close
2016-01-25 · context	49.697	50.377	49.42	49.553
2016-01-26 · context	49.553	50.175	49.324	49.916
2016-01-27 · context	49.763	49.946	48.817	49.009
2016-01-28 · context	49.62	49.956	49.037	49.808
2016-01-29 · context	52.367	52.681	51.668	52.663
2016-02-01 · CUTOFF	52.51	52.711	52.142	52.347

AAPL-2016-03 — context and cutoff only (archive-adjusted USD)

Archive date / clock	Open	High	Low	Close
2016-02-23 · context	93.314	93.41	91.523	91.659
2016-02-24 · context	90.97	93.294	90.331	93.022
2016-02-25 · context	92.975	93.66	92.202	93.66
2016-02-26 · context	94.087	94.884	93.488	93.808
2016-02-29 · context	93.758	95.085	93.555	93.594
2016-03-01 · CUTOFF	94.522	97.542	94.299	97.309

SPY-2016-10 — context and cutoff only (archive-adjusted USD)

Archive date / clock	Open	High	Low	Close
2016-09-26 · context	210.76	210.97	209.77	209.99
2016-09-27 · context	209.81	211.41	209.39	211.3
2016-09-28 · context	211.55	212.53	210.46	212.35
2016-09-29 · context	212.11	212.58	209.8	210.43
2016-09-30 · context	211.38	212.82	211.1	212.01
2016-10-03 · CUTOFF	211.54	211.75	210.78	211.5

Compare the continuation with your frozen note. This reference includes outcome headings, complete ledgers and solutions. In the learner print edition it starts on a new page after the cutoff packet.

Read a case in three passes

First read the information cutoff and cover the continuation. State what is observed, what is hypothesized and what action is allowed. Second reveal the next observation and apply the frozen rule, including a skip or an unresolved state when appropriate. Third reconcile the result and ask which conclusion it supports. This procedure turns a case from a memorable story into something another reader can reproduce.

The first seven cases below are fictional, purpose-built to exercise different decisions. The historical additions are separately labeled: sparse SEC-reported GME facts, Apple's split announcement, and a calendar-selected AAPL/MSFT/SPY gallery with exact archived OHLC. The gallery preserves its complete selection ledger and a separately disclosed ambiguity diagnostic. Each source has a stated data/reuse contract. Hypothetical fills in the fictional records are not historical transactions, and the historical path labels do not estimate strategy performance.

Use the completed P1 plan from Chapter 21 for the shared breakout cases. For ATLS, BRDG, SKIP and AMB, the boundary is $50, S0 closes $50.40, the rounded ceiling is $50.90, the maximum order is 50 shares and the fixed stop/target are $48.60/$54.60. The opening gate, first-hour expiry, partial-fill handling, stop-first convention, ten-session time limit and $2.50-per-side fee all remain P1. ATLS's calendar/account inputs are the same as BRDG's. Each case starts in its own $20,000 cash account unless noted; never add these selected outcomes into one claimed portfolio history.

Case one: follow-through and an orderly failure share a beginning

At S0 the only price-based signal fact is C50.40 > previously frozen 50.00. In both branches the opening offer is $50.60 and a supplied paper execution confirms 50 shares at that price during P1's allowed hour. The entry debit is $2,532.50 including its $2.50 charge; 1R freezes at $100. Read the OHLC table one session at a time. Neither branch can borrow the other's outcome to justify the initial decision.

In ATLS, S2 closes $52 after a low 50.70, S3 closes 52.80, and S4 opens 53.80 and trades to 54.80. Its low 52.80 is above the stop, and its high is strictly beyond the target, so the model

exits at 54.60. Exit credit $2,727.50 brings cash to $20,195. Gross $200 less $5 costs gives $195/+1.95R. This supplies a successful implementation under the model, not a frequency of target attainment.

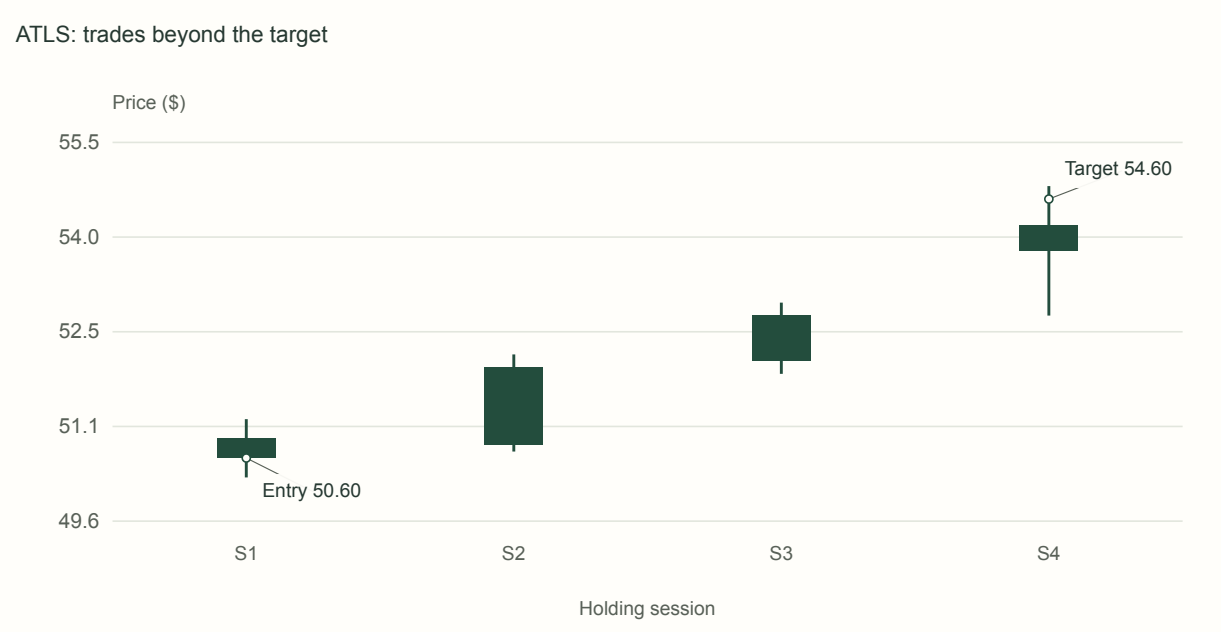
BRDG follows the failure record already completed in the plan, note and journal. S2's close 49.60 weakens the range-continuation interpretation, while its low 49.40 remains above the operational stop. S3 low 48.40 crosses 48.60; the exact-stop model exits at 48.60, yielding –$100 gross, –$105 net and cash $19,895. Rewriting the exit to S2 close after seeing S3 would evaluate a new rule on its discovery case. Both branches follow the initial protocol; their outcomes differ.

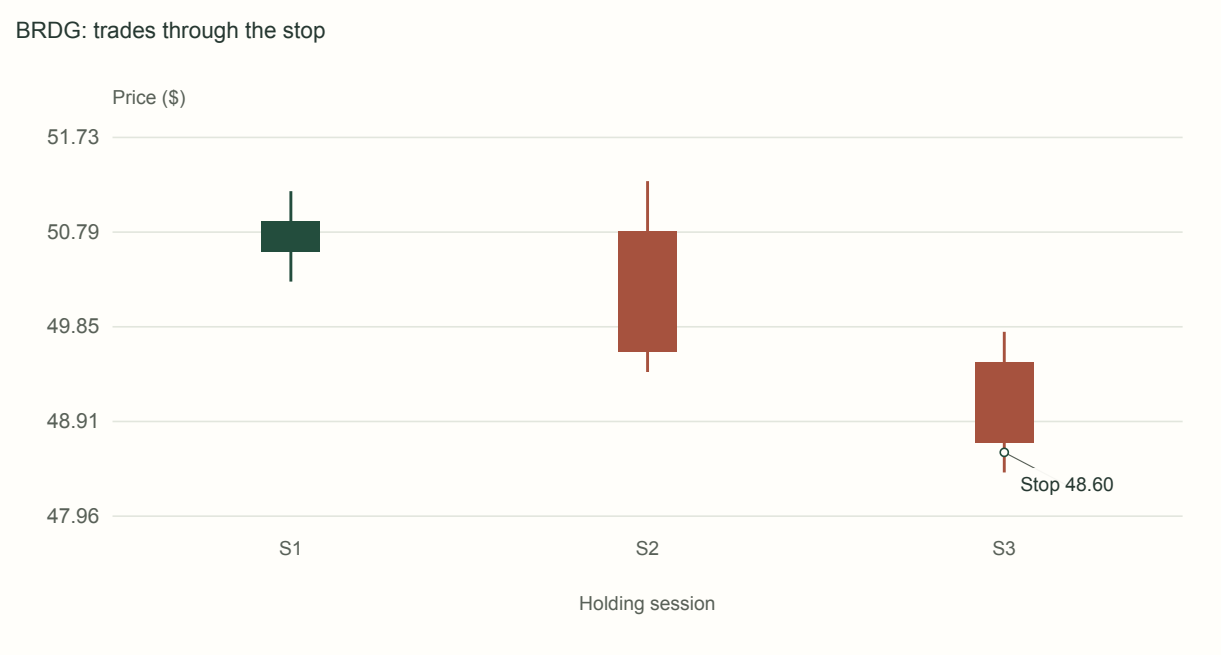
Two P1 continuations from the same entry (OHLC in dollars)

Holding session	ATLS favorable O/H/L/C ($)	BRDG failure O/H/L/C ($)
S1	50.60 / 51.20 / 50.30 / 50.90	Same bar
S2	50.80 / 52.20 / 50.70 / 52.00	50.80 / 51.30 / 49.40 / 49.60
S3	52.10 / 53.00 / 51.90 / 52.80	49.50 / 49.80 / 48.40 / 48.70
S4	53.80 / 54.80 / 52.80 / 54.20	Already exited under P1

FICTIONAL TEACHING ILLUSTRATION

One frozen P1 entry, two observed continuations





Fictional OHLC from the case table, in dollars with separately autoscaled panels. S1 entry is supplied by the paper report; later exit prices use P1's explicit fill model.

▼ View the illustration's data

ATLS: trades beyond the target

Interval	Open	High	Low	Close
S1	50.6	51.2	50.3	50.9
S2	50.8	52.2	50.7	52
S3	52.1	53	51.9	52.8
S4	53.8	54.8	52.8	54.2

BRDG: trades through the stop

Interval	Open	High	Low	Close
S1	50.6	51.2	50.3	50.9
S2	50.8	51.3	49.4	49.6
S3	49.5	49.8	48.4	48.7

Case two: a gap passes the protective price

GAPS is a different price-scale case, with a complete smaller rule GR1. At G0, the supplied close is $79.80 above a previously mapped $79.00 boundary; the calendar has no known earnings through G10. The next-opening gate accepts prices above the $77 fixed stop and at or below $80.60. At the supplied G1 open of $80, a paper report fills 30 shares. Freeze 1R = 30 × ($80 − $77) = $90. Target is $86, time exit G10 close, and the same earlier-holding exit priorities and $5 round-trip fee apply.

The entry session has O80/H81/L79/C80.50, so neither level executes. Before G2, unexpected adverse news arrives. G2 has O70/H72/L68/C71, and the opening-gap priority exits at 70. Price loss is 30 × ($70 − $80) = −$300; net −$305; net R = −305 ÷ 90 = −3.3889. Entry debit $2,402.50 and exit credit $2,097.50 reconcile $19,695 final cash. Scheduled-event eligibility did not exclude the surprise.

Reduce quantity to 15 without changing prices or the flat fees. Planned price-risk 1R becomes $45, notional $1,200 and price loss $150, still −3.3333R gross. Net loss is $155, or −3.4444R: halving quantity reduces dollars, but the fixed fee is now a larger fraction of R. If one instead assumed proportional fees, net R could remain unchanged. Name that cost convention before making the comparison; neither smaller sizing nor a different order makes the $77 stop a price guarantee.

Case three: the skip is a completed decision

SKIP has P1’s S0 signal and all eligibility inputs, but the supplied S1 opening offer is $52. It exceeds 50.90, so the gate rejects the candidate. No order, position or fee exists. A later S2 close of $56 is revealed only after the decision. Record “opening ceiling; no trade,” not a zero-R scratch. Without a position there is no trade-specific denominator.

The missed advance is observable, and it can motivate a ceiling-change hypothesis. It cannot by itself price the alternative strategy. Buying at 52 and selling at 56 would require two specific, feasible fill assumptions; the later close is not a promise that an order could have filled there. A complete comparison would also keep the high-opening candidates that later fell. This case is retained in Chapter 23’s journal so the opportunity restriction remains inspectable.

Case four: two clocks and a daily ambiguity

MIXD isolates a difference between two clocks without an order. At M0, the latest two completed weekly closes are $48.90 and $49.90, up under Chapter 17’s close-direction

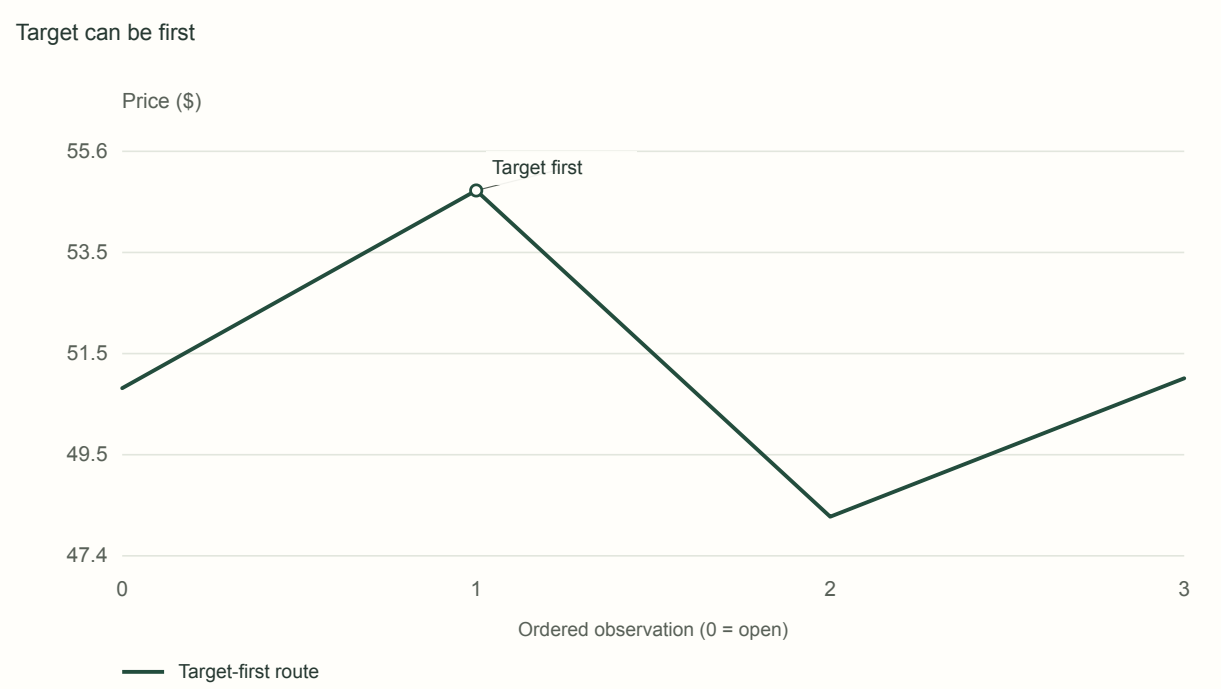
description. M0 is the first session of a new week, and its daily close is $49.70: down from $49.90 and below a previously frozen $50 boundary. Chapter 17 would give this down day neither the aligned nor the conflicting label, because those labels apply only to upward daily changes; the two clocks simply describe different intervals. Rule M1 uses completed-week direction only as context and requires a daily close strictly above 50 before any next-opening order. M0 supplies no daily signal, so no trade is authorized. Calling the rising weekly context "confirmation" cannot supply the missing daily trigger.

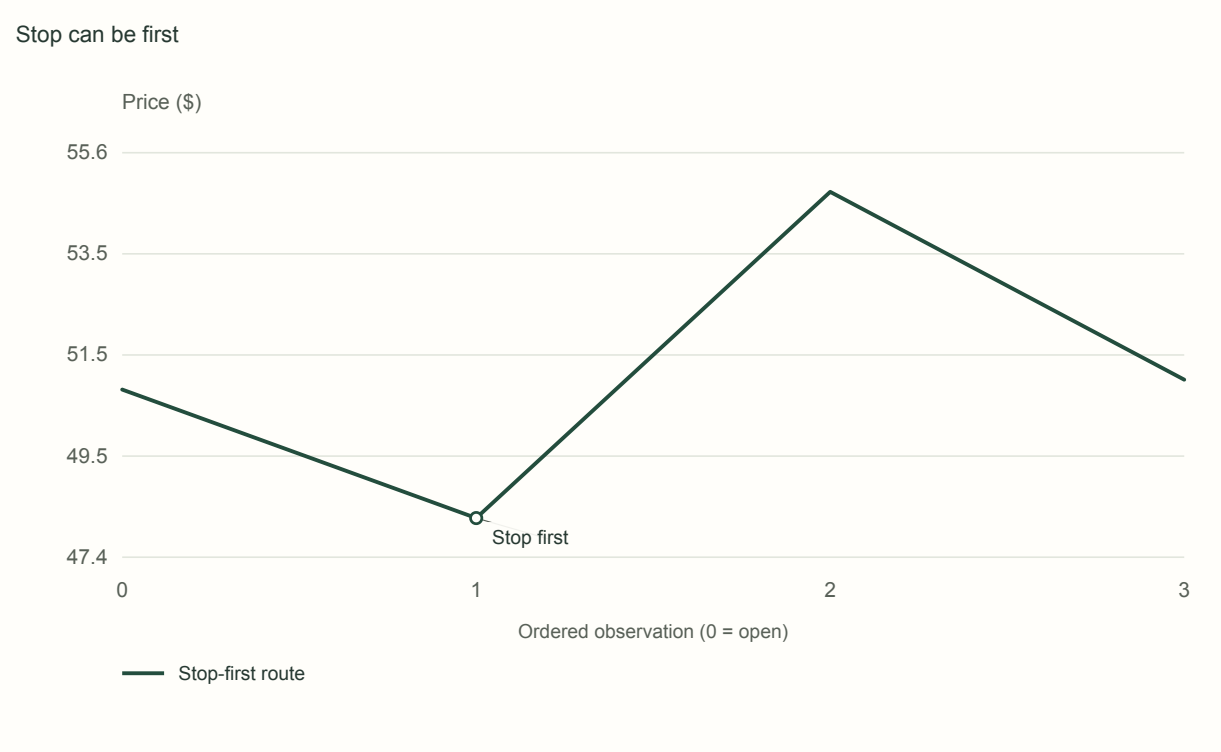
AMB is a separate P1 filled-position branch. At its decision cutoff, it shares the 50-share 50.60 entry, $100 price-risk 1R, stop 48.60 and target 54.60. Its next held-position bar is O50.80/H54.80/L48.20/C51.00. Both stop and target are reached, and the open lies between them. P1 records a stop-first modeled loss of $105/1.05R. A target-first model would give +$195/+1.95R: a $300 or 3R difference caused entirely by a sequence assumption.

Two ordered routes demonstrate what daily OHLC loses: 50.80→54.80→48.20→51.00 can reach the target first; 50.80→48.20→54.80→51.00 can reach the stop first. They share identical OHLC. The daily bar does not select one. The honest record therefore has both a scored MODEL result under P1 and an unresolved true sequence. If no sequence policy had been declared, the case could not be assigned a unique modeled result either.

FICTIONAL TEACHING ILLUSTRATION

AMB: identical OHLC, different first exits





Fictional ordered observations with identical O50.80/H54.80/L48.20/C51.00. These two feasible routes illustrate the unresolved sequence; they do not supply actual intraday data.

▼ View the illustration's data

Target can be first

Ordered observation (0 = open)	Target-first route
0	50.8
1	54.8
2	48.2
3	51

Stop can be first

Ordered observation (0 = open)	Stop-first route
0	50.8
1	48.2
2	54.8
3	51

Case five: a range recovery uses the same decision discipline

RIVR adds a setup beyond breakout continuation. At R0, a supplied earlier range runs from $30 to $34. Its support 30 is frozen before a breach bar O30.50/H30.60/L29.50/C29.80. At that close, the only record is “breach; recovery not yet seen.” As in Chapter 27, a breach is a completed session whose low is at least 1% below support ($29.70 or lower). Rule RR1 opens one recovery window, from the breach session’s own close through the next three sessions; later lows inside that window belong to the same breach event and do not restart it. The first close in the window strictly above 30 and no higher than 34 is the recovery trigger. It is a defined range-recovery study, not proof of hidden accumulation or a complete Wyckoff analysis.

At R1 the stock has O29.90/H30.80/L29.70/C30.60. R1’s own $29.70 low belongs to the open R0 breach event rather than starting a new one, and its completed close meets RR1’s trigger. Freeze an entry rule for R2: accept the next open only in ($29.00, $31.00], buy 50 shares at that supplied opening execution, then use a fixed $29 stop and $35 target, R11 close time exit (ten holding sessions from R2), P1’s stop-first conflict policy and $5 round-trip costs. Account and calendar are clear through R11. Quantity is fixed but price-risk varies with the actual fill; an opening price above $31.00 would instead be a skip.

The accepted R2 bar is O31.00/H32.20/L30.80/C32.00, and a supplied paper report fills 50 at 31.00. Price-risk 1R = 50 × (31−29) = $100; the completed R1 signal did not justify a buy at its earlier 29.90 open. The R3 bar is O32.10/H33.80/L31.80/C33.50. R4 is O34.00/H35.20/L33.70/C34.90, trading strictly above 35 without reaching 29, so the model exits at 35. Gross is $200, net $195/+1.95R, ending cash $20,195. Choosing a $31 ceiling is an arbitrary teaching setting, written at R1, not selected for this ending.

A failed alternative shares R0–R2, then gives R3 O30.70/H31.20/L28.80/C29.20. The modeled 29 stop yields −$100 gross, −$105 net and −1.05R. A missing R1 close would leave recovery eligibility unresolved and create no order. A third branch opens 31.20 at R2, above the 31 ceiling, and is skipped even if it later reaches 35. Keep all such dispositions when designing a study; a recovery label does not guarantee a recovery trade.

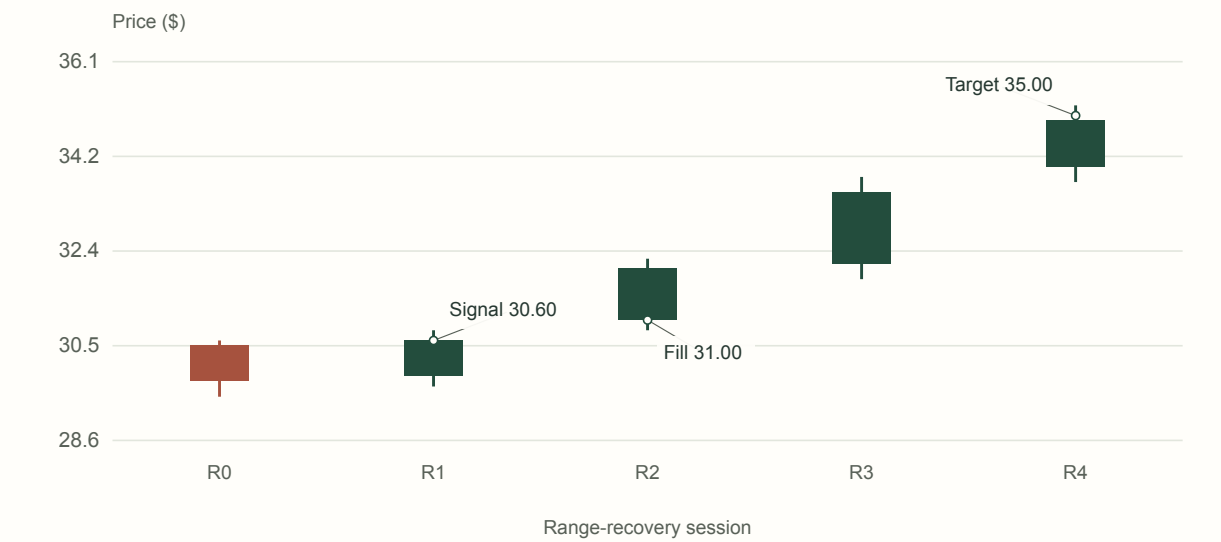
RIVR supplied states: reveal in order

Session	Accepted-path O/H/L/C ($)	Record
R0	30.50 / 30.60 / 29.50 / 29.80	Breach; no recovery trigger
R1	29.90 / 30.80 / 29.70 / 30.60	R1 close: recovery trigger; freeze R2 entry ceiling 31

Session	Accepted-path O/H/L/C ($)	Record
R2	31.00 / 32.20 / 30.80 / 32.00	Supplied fill 50 @31; stop 29/target 35; 1R100
R3	32.10 / 33.80 / 31.80 / 33.50	Neither exit
R4	34.00 / 35.20 / 33.70 / 34.90	Target model 35; net +195

FICTIONAL TEACHING ILLUSTRATION

RIVR: breach, recognition, execution and target



Fictional bars from the range-recovery table. Recognition occurs at R1 close; the supplied execution is R2 open 31.00, not an earlier favorable price.

▼ View the illustration's data

RIVR: breach, recognition, execution and target

Interval	Open	High	Low	Close
R0	30.5	30.6	29.5	29.8
R1	29.9	30.8	29.7	30.6
R2	31	32.2	30.8	32
R3	32.1	33.8	31.8	33.5
R4	34	35.2	33.7	34.9

Compare records without claiming a strategy sample

The completed summary distinguishes a trade's net P&L from a skip and an unresolved execution fact. Each filled result reports its own frozen price-risk denominator and its separate fee model. The gap uses a different price scale and stop distance; the recovery uses a different trigger. Summing their R as if they were interchangeable observations from one locked rule would obscure the very distinctions the cases teach.

Selection explains what this casebook can support. We deliberately chose follow-through, ordinary loss, gap, skip, two-clock and recovery paths. Those cases demonstrate rule application; they were not sampled from a real opportunity set. A count of winners in the gallery would mostly measure the author's selection. Chapter 24 shows how a frozen research rule and chronological evaluation window address a different question.

The historical additions that follow have distinct evidence contracts. Sparse source facts teach when an execution claim must stop; the calendar-selected archive gallery supplies small OHLC windows with retrospective adjustments and undocumented session details. Apply the same three-pass method: freeze the stated clock, identify missing inputs and distinguish later classifications from the earlier decision. Neither documented market facts nor a systematically recorded gallery turn a hypothetical fill into a historical transaction.

Completed case ledger (separate teaching accounts)

Case / rule	Filled shares	Frozen price-risk 1R ($)	Gross ($)	Costs ($)	Net ($)	Net R / disposition
ATLS / P1	50	100	200	5	195	+1.95R
BRDG / P1	50	100	-100	5	-105	–1.05R
GAPS / GR1	30	90	-300	5	-305	–3.3889R
SKIP / P1	0	N/A	N/A	0	N/A	Opening gate rejected; cash unchanged
MIXD / M1	0	N/A	N/A	0	N/A	Daily trigger absent; context not permission
AMB / P1	50	100	-100	5	-105	–1.05R under stop-first; true sequence unresolved
RIVR / RR1	50	100	200	5	195	+1.95R

Case / rule	Filled shares	Frozen price-risk 1R ($)	Gross ($)	Costs ($)	Net ($)	Net R / disposition
RIVR failed branch / RR1	50	100	-100	5	-105	–1.05R; alternative path, not extra sample

Historical case selection and data conventions

The historical cases below were selected to teach two data problems: an event-driven price move and a share-unit change. Selection is purposive, based on publicly documented episodes, rather than a random screen or an outcome-blind trading sample. Both the surge and subsequent fall in the selected GME episode are included. These cases cannot estimate a hit rate. Use the fictional cases earlier in the chapter to rehearse fully specified execution; these historical records teach what can and cannot be inferred from limited observations.

Provenance: GME, January 12–28, 2021, is reconstructed from SEC staff report section 3.2, printed pp. 18–19, published October 14, 2021. Prices use the dollar/share units reported for that period, not a later split-adjusted chart. Reported closing prices are distinguished from intraday highs; the report does not supply a complete OHLC series or exact timestamps for the highs. No dividend reinvestment or price adjustment is performed here. Two closes are derived from explicitly reported daily dollar changes, with the arithmetic disclosed in the table. Missing sessions remain missing.

Reuse basis: the SEC's Webmaster FAQ states that Government-created site content is free to access and reuse, while some content such as stock art is excluded. This book attributes a few numeric facts from the staff report and makes its own table and diagram; it copies no vendor chart, artwork or logo. The FAQ is not a license for a vendor's underlying market-data feed. The Apple case paraphrases the issuer's dated announcement and uses original algebra, with no copied chart or photo.

GME event reaction: a large move still leaves execution unknown

At the January 13 close, compare the two completed closing observations and volumes shown in the first panel. Price change is 31.40 − 19.95 = $11.45, or 57.39%; approximate share volume is roughly 20 times the previous session (about 144 million against 7 million shares). These are observations. The words "breakout" or "short squeeze" add an interpretation whose definition and evidence must be supplied.

Freeze a retrospective teaching decision at that close: require enough preceding OHLC to map and freeze a range boundary before January 13 (P1's signal input), a defined event calendar, and an executable next-session opening offer before accepting P1's long setup. The extracted report supplies none of those three inputs. The decision is "insufficient data; no modeled order." January 27's later price is hidden from this decision. A move eventually reaching a much higher price cannot repair missing information at the earlier cutoff.

DOCUMENTED HISTORICAL DATA

GME: sparse documented observations, with cutoffs preserved

Two-session reaction known at January 13 close

U.S. session	Close ($/share)	Share volume (approx.)
2021-01-12	19.95	7 million
2021-01-13	31.4	144 million

Three consecutive closes, read in chronological order

U.S. session	Close ($/share)	Provenance
2021-01-26	147.98	347.51 − 199.53 (derived)
2021-01-27	347.51	Reported close
2021-01-28	193.6	347.51 − 153.91 (derived)

Different observation type: not a closing fill

U.S. session	Intraday high ($/share)
2021-01-28	483

SEC staff report (2021), section 3.2, pp. 18–19. Original table of reported facts and explicitly derived closes; omitted sessions are not interpolated. The intraday high is separate from the closes.

▼ View the illustration's data

Two-session reaction known at January 13 close

U.S. session	Close ($/share)	Share volume (approx.)
2021-01-12	19.95	7 million
2021-01-13	31.4	144 million

Three consecutive closes, read in chronological order

U.S. session	Close ($/share)	Provenance
2021-01-26	147.98	347.51 – 199.53 (derived)
2021-01-27	347.51	Reported close
2021-01-28	193.6	347.51 – 153.91 (derived)

Different observation type: not a closing fill

U.S. session	Intraday high ($/share)
2021-01-28	483

GME surge and reversal: do not turn a high into a closing fill

The second panel contains three consecutive closes. January 26 is $147.98, derived as $347.51 − $199.53; January 28 is $193.60, derived as $347.51 − $153.91. Close-to-close changes are +134.84% then −44.29%. The separately reported January 28 intraday high, $483, is not that day's closing price.

At the January 27 close, a momentum description is supported by the supplied pair of closes. An execution claim is not: the next opening price, spread, halt timing, low, and fill sequence are absent. If a learner writes "bought the January 27 close and sold the January 28 high," mark both fills as invented until independent transaction data supports them. The subsequent fall is a contrasting observation within the same chosen episode, not a countable failure of an unspecified strategy. Review outcome: the direction description transferred; the order model did not have enough input to run.

Apple’s 2014 split: change the unit before classifying a break

Apple announced a seven-for-one stock split on April 23, 2014, with a June 2 record date and split-adjusted trading beginning June 9. This corporate-action case is selected to test unit consistency, independently of any later return. The announcement is usable as an event-calendar fact after its release; it supplies no entry or target prices.

At a June 6 analysis cutoff, the announced unit change belongs in the data plan for the next session. If a pre-split price is P and volume is V, a consistently split-adjusted comparison expresses that prior price as P/7 and prior share volume as 7V. A holder of N shares becomes a holder of 7N shares, so NP = (7N)(P/7) before other price movement. P, V and N are algebraic placeholders, not historical quotes. No dollar P&L is inferred. A raw discontinuity caused by the share unit is not evidence of a structural support failure.

Decision record: request a daily feed with its adjustment convention and the corporate-action flag before drawing a June 9 candle or computing a moving average across the date. The supplied announcement alone leads to a data-check decision, not a trade. This makes “skip” a completed analysis with a reason, rather than an empty chart opinion.

A calendar-selected historical gallery: freeze the procedure first

The next gallery gives you a complete small OHLC window rather than a handful of famous-event facts. Before downloading prices, we fixed three symbols—Apple (AAPL), Microsoft (MSFT) and the SPDR S&P 500 ETF (SPY)—and every month of 2016. The cutoff is the first available archive session on or after each month’s first day. Every one of the 36 symbol-month cells stays in the ledger. The symbols and year are teaching choices, not a historical investable universe; choosing familiar surviving securities introduces a selection limit even when dates are chosen without looking at their outcomes.

The exercise gate is deliberately simple: compare the cutoff close C with the arithmetic mean of the five preceding closes, excluding C. If C is not strictly higher, record “skipped.” If it is higher, observe the next five archive sessions against an upper level of 1.01C and a lower level of 0.99C. A session opening at or beyond a level resolves that side first. Otherwise compare its high and low: a sole upper touch is “favorable,” a sole lower touch “adverse,” and touching both in that first resolving bar is “ambiguous.” No touch within five sessions is “ordinary/unresolved.” These labels describe a price path under an invented exercise rule. They are not successful or failed historical trades.

The full ledger below contains 18 skips, six favorable paths, eleven adverse paths and one ordinary/unresolved path; none is an ambiguous first resolving bar. We show the earliest example of each represented class, with alphabetical symbol order breaking date ties. This representative choice is intentionally outcome-stratified, so the four illustrations are not a sample from which to estimate probabilities. Do not divide favorable counts by eligible counts to infer a strategy's success rate. There is no order, position, fill, stop execution, cash account, commission model or held-out strategy test.

Provenance and reuse: Boris Marjanovic's Huge Stock Market Dataset on Kaggle, version 3, provides the three daily files. Retrieved metadata explicitly declares CC0-1.0, and the depositor states that prices are adjusted for dividends and splits. We copied numeric observations into original tables and diagrams. The stated permission is the depositor's; metadata does not identify an upstream provider or establish others' rights. The local source snapshots, exact download URLs, SHA-256 hashes, extracts and frozen selection sheet are preserved in data/historical/ for independent inspection.

Data conventions: source date labels and all OHLC decimals are retained, in the archive's adjusted dollar scale. Daily aggregation is reported, but exact regular/extended-session inclusion, time zone and adjustment formula are undocumented, and the adjustment vintage is bounded only by the depositor's note "last updated 11/10/2017". We therefore call them archive sessions, not certified regular-session candles. No missing date is interpolated. Volume is unused because its adjustment convention is unspecified. These are retrospectively adjusted records, not a reconstruction of prices published at each 2016 cutoff. The cutoff exercise hides future bars; it does not make the adjusted vintage contemporaneous. Before modeling real transactions, obtain documented raw quotes, corporate actions, precise sessions, spread and order information.

Complete fixed calendar ledger (classification only; no trading performance)

Symbol	2016 cutoff	C / prior-five mean	Path class	First resolving observation
AAPL	2016-01-04	101.44 / 103.238	skipped	No observation gate
MSFT	2016-01-04	52.434 / 53.574	skipped	No observation gate
SPY	2016-01-04	194.03 / 198.476	skipped	No observation gate
AAPL	2016-02-01	92.839 / 93.2486	skipped	No observation gate
MSFT	2016-02-01	52.347 / 50.1898	adverse	2016-02-02 range
SPY	2016-02-01	186.92 / 183.16	adverse	2016-02-02 range
AAPL	2016-03-01	97.309 / 93.1486	favorable	2016-03-03 range
MSFT	2016-03-01	50.671 / 49.4988	adverse	2016-03-03 range

Symbol	2016 cutoff	C / prior-five mean	Path class	First resolving observation
SPY	2016-03-01	191.28 / 187.20	favorable	2016-03-04 range
AAPL	2016-04-01	106.46 / 103.978	favorable	2016-04-04 range
MSFT	2016-04-01	53.552 / 52.5668	adverse	2016-04-04 range
SPY	2016-04-01	200.77 / 198.504	adverse	2016-04-05 open
AAPL	2016-05-02	90.641 / 95.9858	skipped	No observation gate
MSFT	2016-05-02	48.772 / 49.0072	skipped	No observation gate
SPY	2016-05-02	201.78 / 201.944	skipped	No observation gate
AAPL	2016-06-01	95.886 / 97.0214	skipped	No observation gate
MSFT	2016-06-01	51.297 / 50.6428	adverse	2016-06-02 range
SPY	2016-06-01	204 / 203.068	favorable	2016-06-08 range
AAPL	2016-07-01	93.384 / 91.3552	adverse	2016-07-05 range
MSFT	2016-07-01	49.647 / 48.4068	adverse	2016-07-06 range
SPY	2016-07-01	204.66 / 199.362	adverse	2016-07-05 range
AAPL	2016-08-01	103.28 / 98.460	adverse	2016-08-02 range
MSFT	2016-08-01	54.907 / 54.843	favorable	2016-08-04 range
SPY	2016-08-01	211.57 / 211.40	adverse	2016-08-02 range
AAPL	2016-09-01	104.5 / 104.456	favorable	2016-09-02 range
MSFT	2016-09-01	56.225 / 56.5692	skipped	No observation gate
SPY	2016-09-01	212.01 / 212.36	skipped	No observation gate
AAPL	2016-10-03	110.18 / 110.67	skipped	No observation gate
MSFT	2016-10-03	56.069 / 56.2212	skipped	No observation gate
SPY	2016-10-03	211.5 / 211.216	ordinary	No touch in five sessions
AAPL	2016-11-01	109.17 / 112.714	skipped	No observation gate
MSFT	2016-11-01	58.394 / 58.8836	skipped	No observation gate
SPY	2016-11-01	206.83 / 209.006	skipped	No observation gate
AAPL	2016-12-01	107.76 / 109.554	skipped	No observation gate
MSFT	2016-12-01	58.198 / 59.5528	skipped	No observation gate
SPY	2016-12-01	215.22 / 216.426	skipped	No observation gate

AAPL January: a completed skip

At the January 4, 2016 cutoff, the preceding AAPL closes are 104.01, 102.83, 104.68, 103.32 and 101.35. Their sum is 516.19, giving a mean of 103.238. The cutoff close, 101.44, is below that mean, so the frozen exercise records a skip. January 4's own low and high do not count as later level touches: the levels would only be frozen after this candle finishes.

Reveal January 5–11 only after recording the skip. The later decline is consistent with a weak path, but it was not needed to justify the gate decision. A reader who changes the cutoff or adds a short position after seeing the decline has written a different exercise. No skipped-window gain, avoided loss or fill is computed.

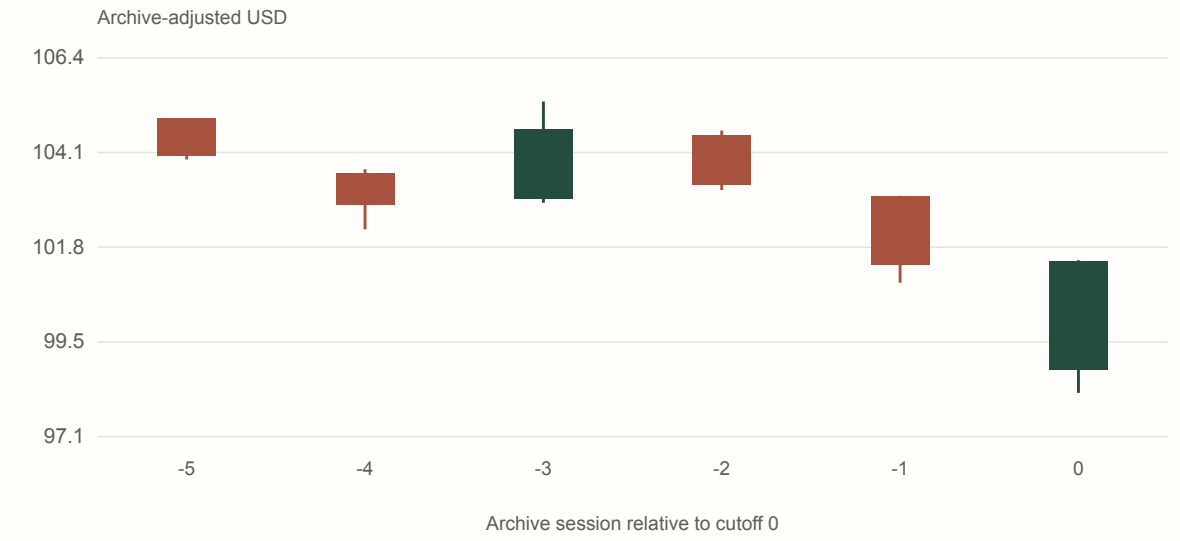
AAPL-2016-01 — exact archived OHLC; full continuation reference; attempt the cutoff-only packet first

Archive date / clock	Open	High	Low	Close
2015-12-24 · context	104.93	104.93	103.93	104.01
2015-12-28 · context	103.59	103.69	102.22	102.83
2015-12-29 · context	102.97	105.35	102.87	104.68
2015-12-30 · context	104.53	104.64	103.18	103.32
2015-12-31 · context	103.02	103.04	100.91	101.35
2016-01-04 · CUTOFF	98.786	101.46	98.211	101.44
2016-01-05 · later	101.81	101.91	98.597	98.885
2016-01-06 · later	96.816	98.558	96.149	96.949
2016-01-07 · later	95.003	96.4	92.839	92.857
2016-01-08 · later	94.879	95.417	93.157	93.349
2016-01-11 · later	95.283	95.369	93.715	94.86

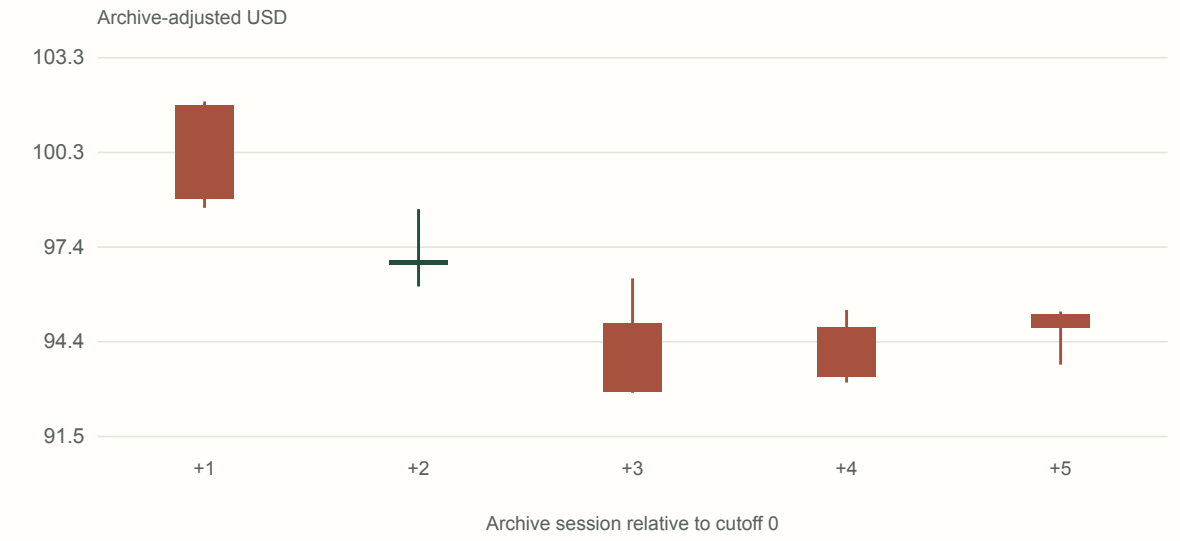
DOCUMENTED HISTORICAL DATA

AAPL January: a completed skip · 2016-01-04

First pass: five preceding bars and cutoff 0



Reveal second: following archive sessions +1 to +5



Historical archive OHLC, Kaggle version 3 (depositor CC0); dates and exact values are in the adjacent table. Retrospectively dividend/split-adjusted scale; exact session and adjustment algorithm undocumented. Panels have separate price axes. Cutoff is bar 0; later bars +1 to +5 are a reveal, not information available at the cutoff. No fills or returns are modeled.

▼ View the illustration's data

First pass: five preceding bars and cutoff 0

Interval	Open	High	Low	Close
-5	104.93	104.93	103.93	104.01
-4	103.59	103.69	102.22	102.83
-3	102.97	105.35	102.87	104.68
-2	104.53	104.64	103.18	103.32
-1	103.02	103.04	100.91	101.35
0	98.786	101.46	98.211	101.44

Reveal second: following archive sessions +1 to +5

Interval	Open	High	Low	Close
+1	101.81	101.91	98.597	98.885
+2	96.816	98.558	96.149	96.949
+3	95.003	96.4	92.839	92.857
+4	94.879	95.417	93.157	93.349
+5	95.283	95.369	93.715	94.86

MSFT February: the continuation check fails

At the February 1, 2016 MSFT cutoff, C = 52.347 exceeds the prior-five mean 50.1898. The gate permits observation; the frozen levels are 52.87047 above and 51.82353 below. This is a continuation hypothesis with a declared falsification check, not a forecast with a known probability.

The next archive session, February 2, opens at 51.830, narrowly inside the levels. Its low 50.377 crosses the lower level while its high 51.917 never reaches the upper one. The first resolving bar therefore classifies adverse. Nearest-cent rounding of 51.82353 gives 51.82: the open 51.830 is above both 51.82353 and 51.82, so the opening classification stays inside in this actual case. Preserve source precision through the comparison. For a separate fictional counterexample, a lower threshold 51.826 rounds to 51.83; an open 51.828 is above the exact threshold but at or below the rounded one. That named rounding policy would change inside to an opening lower-boundary touch. These fictional numbers do not replace the archived historical observations. The subsequent weakness cannot turn the hypothetical threshold into a real stop fill.

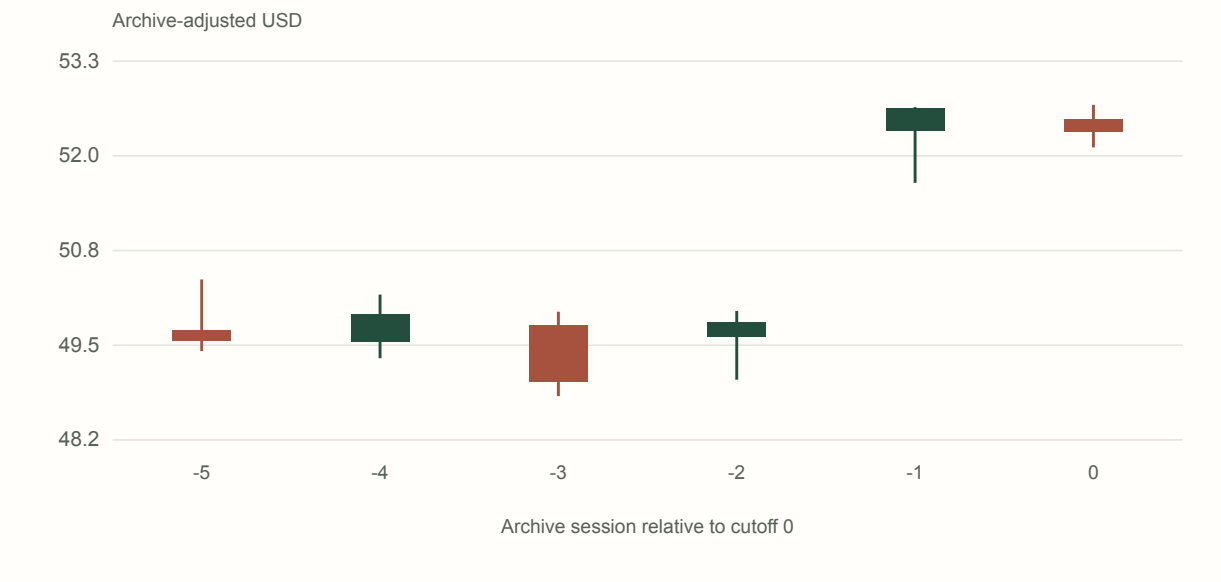
MSFT-2016-02 — exact archived OHLC; full continuation reference; attempt the cutoff-only packet first

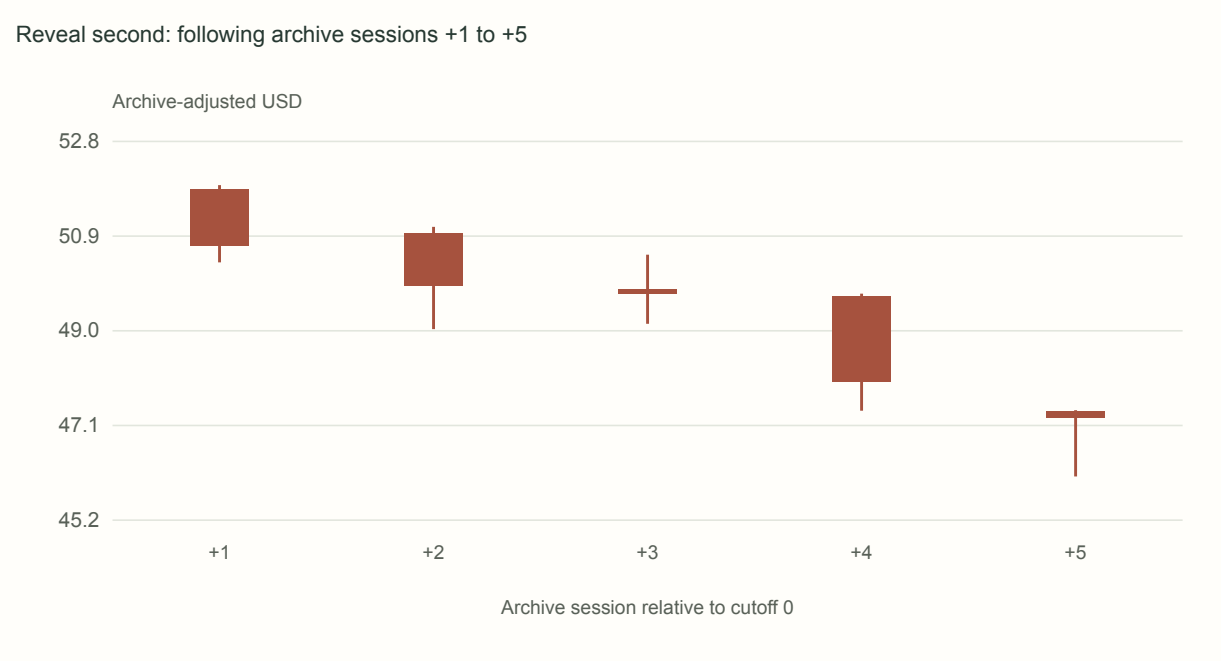
Archive date / clock	Open	High	Low	Close
2016-01-25 · context	49.697	50.377	49.42	49.553
2016-01-26 · context	49.553	50.175	49.324	49.916
2016-01-27 · context	49.763	49.946	48.817	49.009
2016-01-28 · context	49.62	49.956	49.037	49.808
2016-01-29 · context	52.367	52.681	51.668	52.663
2016-02-01 · CUTOFF	52.51	52.711	52.142	52.347
2016-02-02 · later	51.83	51.917	50.377	50.711
2016-02-03 · later	50.951	51.085	49.046	49.906
2016-02-04 · later	49.85	50.53	49.152	49.753
2016-02-05 · later	49.697	49.753	47.419	47.993
2016-02-08 · later	47.409	47.429	46.108	47.276

DOCUMENTED HISTORICAL DATA

MSFT February: the continuation check fails · 2016-02-01

First pass: five preceding bars and cutoff 0





Historical archive OHLC, Kaggle version 3 (depositor CC0); dates and exact values are in the adjacent table. Retrospectively dividend/split-adjusted scale; exact session and adjustment algorithm undocumented. Panels have separate price axes. Cutoff is bar 0; later bars +1 to +5 are a reveal, not information available at the cutoff. No fills or returns are modeled.

▼ View the illustration's data

First pass: five preceding bars and cutoff 0

Interval	Open	High	Low	Close
-5	49.697	50.377	49.42	49.553
-4	49.553	50.175	49.324	49.916
-3	49.763	49.946	48.817	49.009
-2	49.62	49.956	49.037	49.808
-1	52.367	52.681	51.668	52.663
0	52.51	52.711	52.142	52.347

Reveal second: following archive sessions +1 to +5

Interval	Open	High	Low	Close
+1	51.83	51.917	50.377	50.711
+2	50.951	51.085	49.046	49.906
+3	49.85	50.53	49.152	49.753

Interval	Open	High	Low	Close
+4	49.697	49.753	47.419	47.993
+5	47.409	47.429	46.108	47.276

AAPL March: favorable follow-through

At the March 1, 2016 AAPL cutoff, the preceding closes sum to 465.743 and average 93.1486. C = 97.309 passes the gate. The frozen lower/upper levels are 96.33591/98.28209. Write the observation and levels before reading the next five rows.

March 2's entire range, 96.449–97.661, remains between the levels. On March 3, the open 97.357 is also inside, the low 97.232 remains above the lower level, and the high 98.459 exceeds the upper. This is favorable follow-through under the fixed path definition. March 4's later high is irrelevant to the first-touch classification. Calling this a profitable trade would require entry timing, executable quotes, size and costs that the archive does not supply.

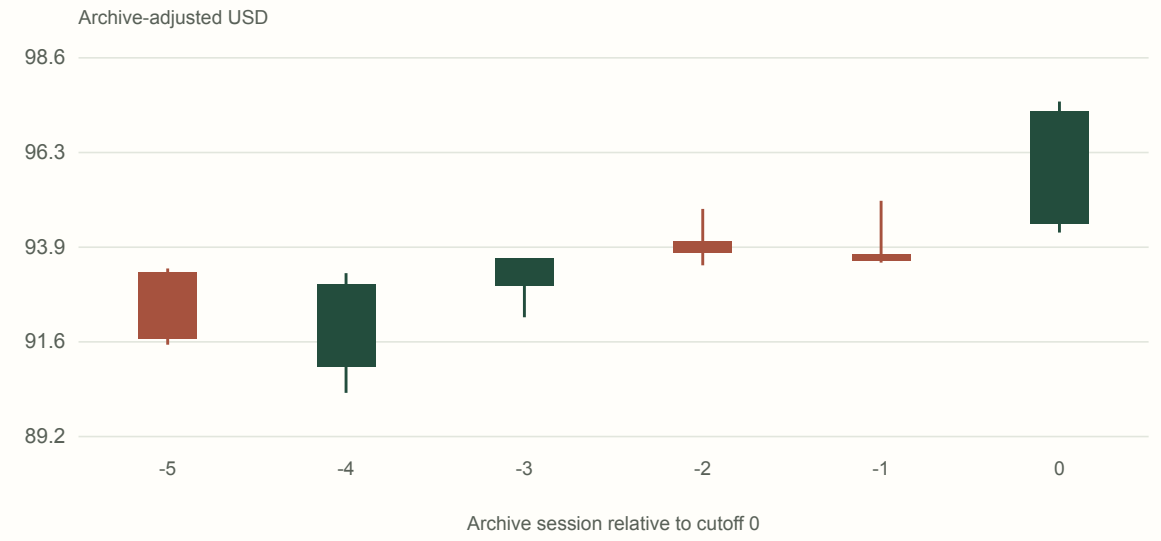
AAPL-2016-03 — exact archived OHLC; full continuation reference; attempt the cutoff-only packet first

Archive date / clock	Open	High	Low	Close
2016-02-23 · context	93.314	93.41	91.523	91.659
2016-02-24 · context	90.97	93.294	90.331	93.022
2016-02-25 · context	92.975	93.66	92.202	93.66
2016-02-26 · context	94.087	94.884	93.488	93.808
2016-02-29 · context	93.758	95.085	93.555	93.594
2016-03-01 · CUTOFF	94.522	97.542	94.299	97.309
2016-03-02 · later	97.289	97.661	96.449	97.522
2016-03-03 · later	97.357	98.459	97.232	98.26
2016-03-04 · later	99.093	100.42	98.132	99.73
2016-03-07 · later	99.113	99.54	97.729	98.607
2016-03-08 · later	97.552	98.508	97.183	97.797

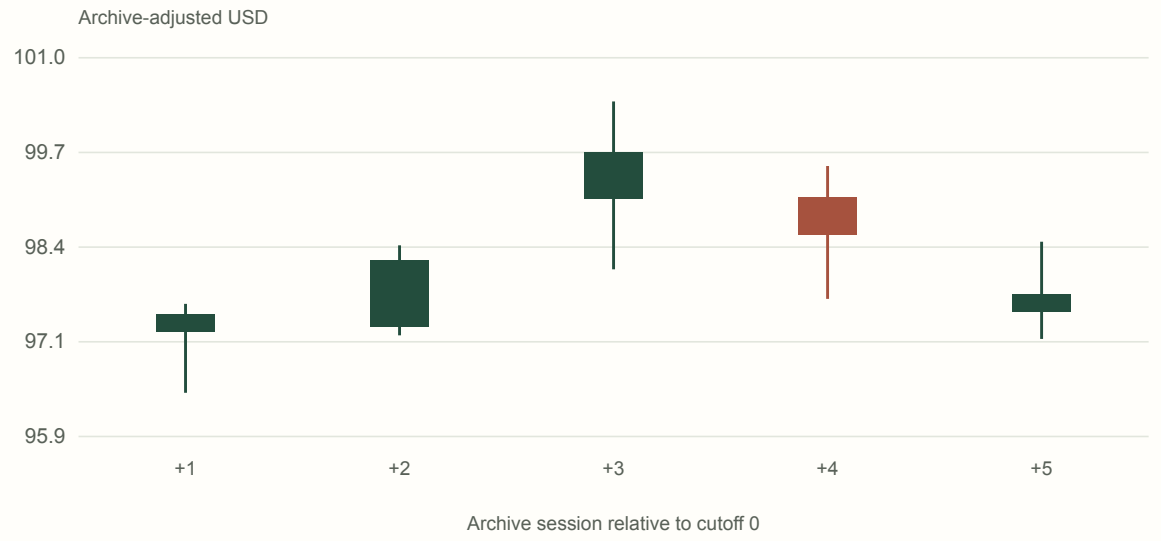
DOCUMENTED HISTORICAL DATA

AAPL March: favorable follow-through · 2016-03-01

First pass: five preceding bars and cutoff 0



Reveal second: following archive sessions +1 to +5



Historical archive OHLC, Kaggle version 3 (depositor CC0); dates and exact values are in the adjacent table. Retrospectively dividend/split-adjusted scale; exact session and adjustment algorithm undocumented. Panels have separate price axes. Cutoff is bar 0; later bars +1 to +5 are a reveal, not information available at the cutoff. No fills or returns are modeled.

▼ View the illustration's data

First pass: five preceding bars and cutoff 0

Interval	Open	High	Low	Close
-5	93.314	93.41	91.523	91.659
-4	90.97	93.294	90.331	93.022
-3	92.975	93.66	92.202	93.66
-2	94.087	94.884	93.488	93.808
-1	93.758	95.085	93.555	93.594
0	94.522	97.542	94.299	97.309

Reveal second: following archive sessions +1 to +5

Interval	Open	High	Low	Close
+1	97.289	97.661	96.449	97.522
+2	97.357	98.459	97.232	98.26
+3	99.093	100.42	98.132	99.73
+4	99.113	99.54	97.729	98.607
+5	97.552	98.508	97.183	97.797

SPY October: an ordinary unresolved window

At the October 3, 2016 SPY cutoff, C = 211.500 exceeds the prior-five mean 211.216. Its 209.385/213.615 levels are therefore observed over October 4–10. Across those five sessions, the lowest low is 209.750 and the highest high is 212.410. Both are inside the levels; no open can have crossed a level while lying within those recorded daily extremes.

The record ends ordinary/unresolved, even though individual closes move and the final close is slightly higher than the cutoff. A binary win/loss story would erase this case. Extending the horizon until something dramatic happens would revise the frozen observation window. The everyday lesson is that a completed analysis can reach its time limit without resolving the proposed interpretation.

SPY-2016-10 — exact archived OHLC; full continuation reference; attempt the cutoff-only packet first

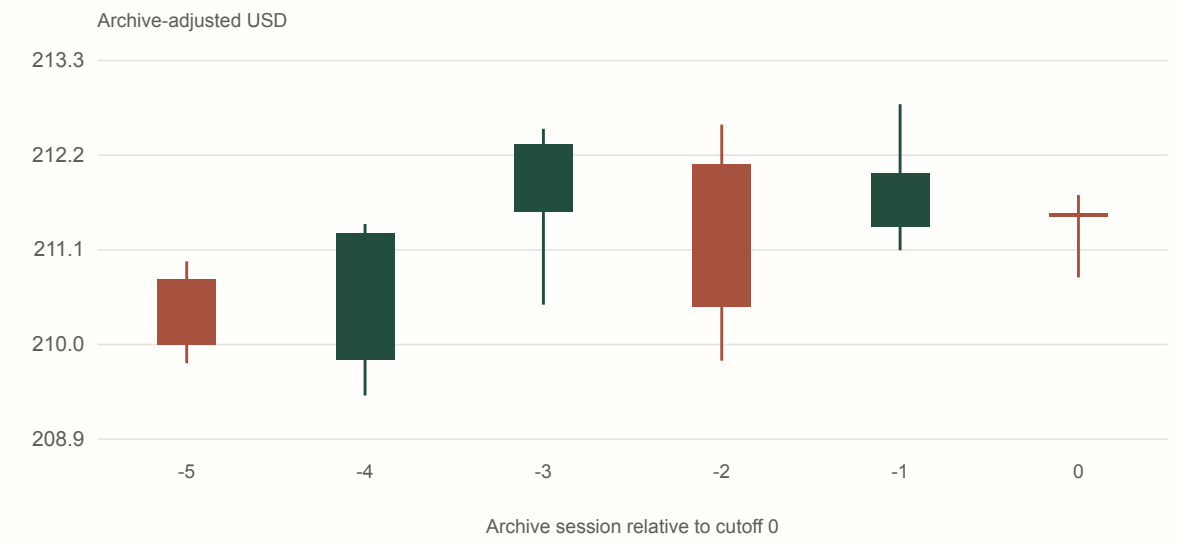
Archive date / clock	Open	High	Low	Close
2016-09-26 · context	210.76	210.97	209.77	209.99

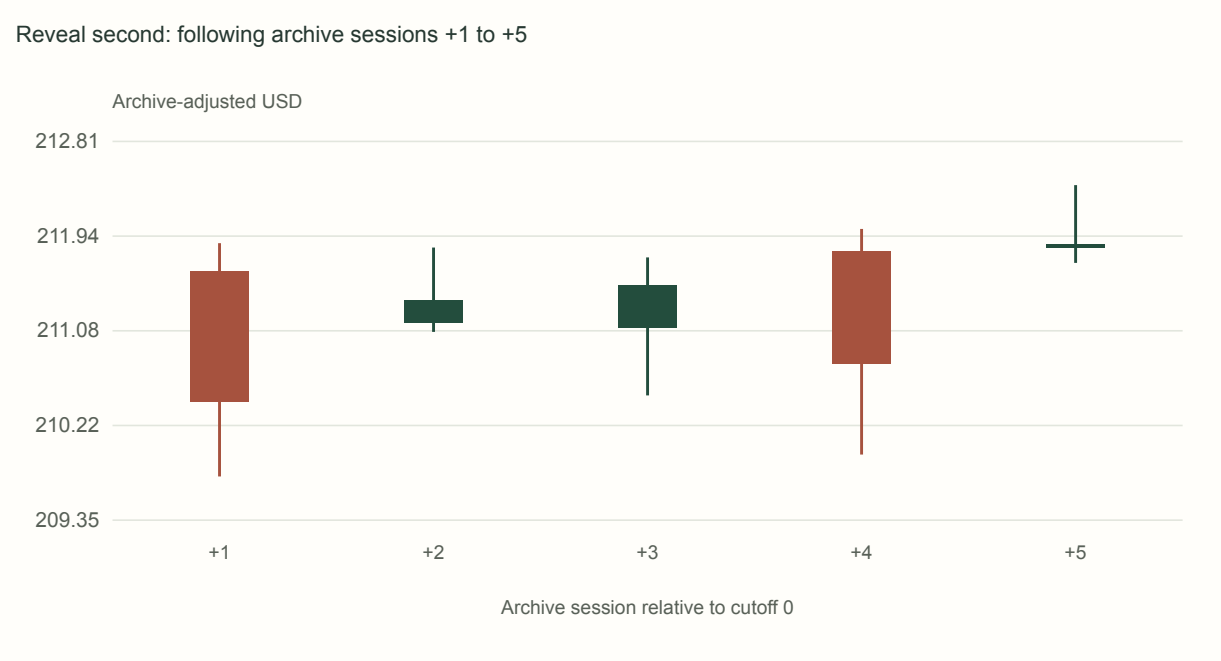
Archive date / clock	Open	High	Low	Close
2016-09-27 · context	209.81	211.41	209.39	211.3
2016-09-28 · context	211.55	212.53	210.46	212.35
2016-09-29 · context	212.11	212.58	209.8	210.43
2016-09-30 · context	211.38	212.82	211.1	212.01
2016-10-03 · CUTOFF	211.54	211.75	210.78	211.5
2016-10-04 · later	211.62	211.88	209.75	210.43
2016-10-05 · later	211.15	211.84	211.07	211.36
2016-10-06 · later	211.11	211.75	210.49	211.5
2016-10-07 · later	211.81	212.01	209.95	210.78
2016-10-10 · later	211.87	212.41	211.7	211.87

DOCUMENTED HISTORICAL DATA

SPY October: an ordinary unresolved window · 2016-10-03

First pass: five preceding bars and cutoff 0





Historical archive OHLC, Kaggle version 3 (depositor CC0); dates and exact values are in the adjacent table. Retrospectively dividend/split-adjusted scale; exact session and adjustment algorithm undocumented. Panels have separate price axes. Cutoff is bar 0; later bars +1 to +5 are a reveal, not information available at the cutoff. No fills or returns are modeled.

▼ View the illustration's data

First pass: five preceding bars and cutoff 0

Interval	Open	High	Low	Close
-5	210.76	210.97	209.77	209.99
-4	209.81	211.41	209.39	211.3
-3	211.55	212.53	210.46	212.35
-2	212.11	212.58	209.8	210.43
-1	211.38	212.82	211.1	212.01
0	211.54	211.75	210.78	211.5

Reveal second: following archive sessions +1 to +5

Interval	Open	High	Low	Close
+1	211.62	211.88	209.75	210.43
+2	211.15	211.84	211.07	211.36
+3	211.11	211.75	210.49	211.5

Interval	Open	High	Low	Close
+4	211.81	212.01	209.95	210.78
+5	211.87	212.41	211.7	211.87

AAPL February: an ambiguous candle inside a skipped window

The fixed ledger has no ambiguous first resolving bar. We did not change its rule or add dates to manufacture one. After reviewing the cohort, we separately looked for a later candle whose open lies between the frozen levels while its high and low span both. The first such date is February 5 in the AAPL February window. This is a disclosed post-audit diagnostic, selected for its geometry; it does not replace that window's skipped classification or establish an ambiguous trade.

At the February 1 cutoff, C = 92.839 is below the prior-five mean 93.2486, so the original gate skips. Its analytical levels, retained only for this diagnostic, are 91.91061 and 93.76739. Pose a fresh question after February 4 closes: what can the next daily candle tell us about the order in which those two levels are visited? No position is assumed to survive earlier bars, and neither later extreme is known at that question cutoff.

Reveal February 5: O/H/L/C = 93.429/93.816/90.690/91.009. The open is between the levels, high is above the upper, and low below the lower. Both high-before-low and low-before-high are compatible with these four observations. The alternative lines below are invented possible orders through the recorded extrema, not measured intraday paths or times. A target-first or stop-first assertion needs intraday observations and an execution model; a daily bar alone cannot settle it. Keep this ambiguity visible without undoing the original skip. One further caution concerns the data themselves. Apple's $0.52 quarterly dividend was payable to holders of record on February 8, so under the 2016 settlement cycle the shares went ex-dividend on February 4, between the February 1 levels and this bar. The archive adjusts for dividends, so the levels and the bar carry different adjustment factors, and the high clears the adjusted upper level by only about $0.05, far less than the dividend. In unadjusted prices this bar probably reached only the lower level. The lesson concerns what any daily bar cannot reveal about order; it is not a claim that published February 5 prices spanned both levels.

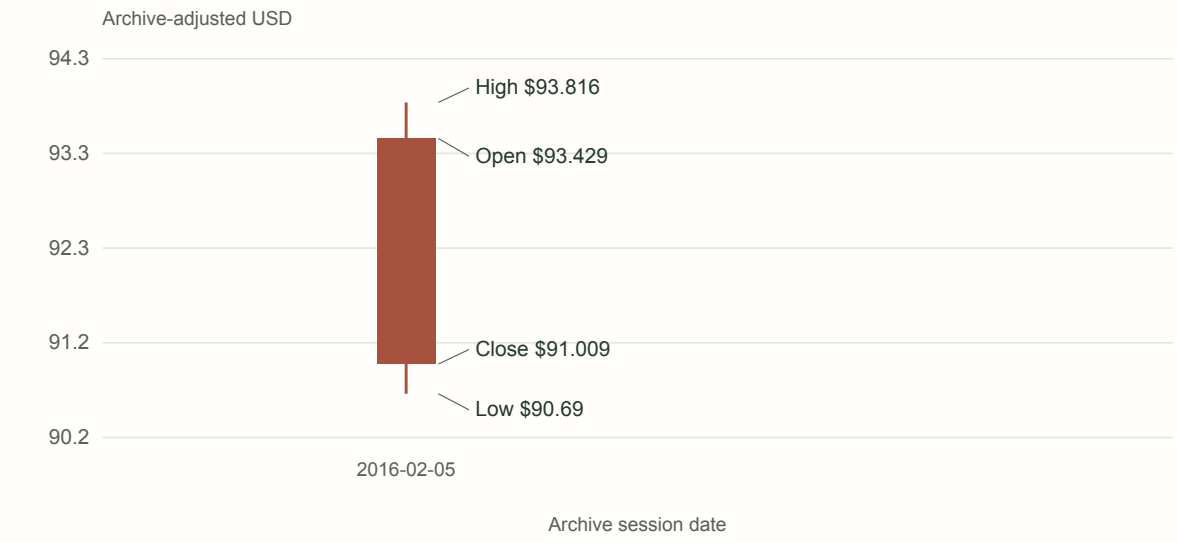
Skipped February window: cutoff and later diagnostic inputs (archive-adjusted USD)

Archive date / clock	Open	High	Low	Close
2016-01-25	97.739	97.748	95.517	95.736
2016-01-26	96.209	97.124	94.418	96.266
2016-01-27	92.463	93.029	89.863	89.941
2016-01-28	90.295	91	88.948	90.585
2016-01-29	91.258	93.715	90.835	93.715
2016-02-01 · original cutoff	92.876	93.108	91.847	92.839
2016-02-02	91.867	92.463	90.769	90.961
2016-02-03	91.462	93.233	90.575	92.761
2016-02-04 · new question cutoff	92.79	94.214	92.142	93.506
2016-02-05 · diagnostic reveal	93.429	93.816	90.69	91.009
2016-02-08	90.148	92.636	90.06	91.968

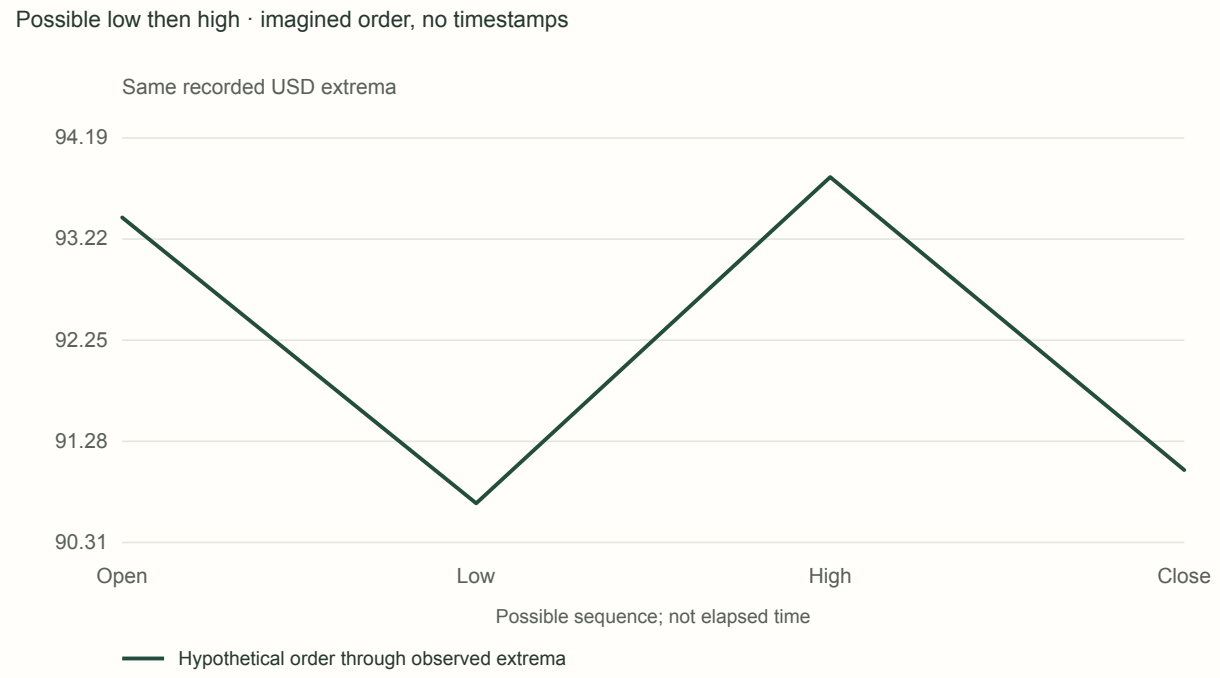
DOCUMENTED HISTORICAL DATA

One historical candle, two possible orders

Actual archived daily bar · February 5, 2016







AAPL, February 5, 2016; archive version 3, depositor CC0. The first panel is historical OHLC; the two lower panels are hypothetical sequences using those extrema, not intraday measurements. This post-audit geometry example comes from a skipped window and is excluded from first-touch outcome counts. A dividend went ex on February 4, so in unadjusted prices the bar may not have reached the upper level.

▼ View the illustration's data

Actual archived daily bar · February 5, 2016

Interval	Open	High	Low	Close
2016-02-05	93.429	93.816	90.69	91.009

Possible high then low · imagined order, no timestamps

Possible sequence; not elapsed time	Hypothetical order through observed extrema
Open	93.429
High	93.816
Low	90.69
Close	91.009

Possible low then high · imagined order, no timestamps

Possible sequence; not elapsed time	Hypothetical order through observed extrema
Open	93.429
Low	90.69
High	93.816
Close	91.009

A WORKED EXAMPLE

A fee convention changes the smaller gap's net R

GAPS, 30 shares: 1R $90, gross loss $300, costs $5, net −$305/−3.3889R. At 15 shares, 1R $45, gross loss $150, costs still $5, net −$155/−3.4444R. Gross R stays −3.3333; net R worsens slightly because the flat fee is a larger share of planned risk.

Pause & try it

Recalculate GAPS at 15 rather than 30 shares, including the $5 flat round-trip cost. Give price-risk 1R, notional, gross/net loss, net R, cash after entry and final cash. Explain what changes and

what remains constant.

GUIDED PRACTICE

Run the recovery before viewing R3

At R1 close 30.60, apply the frozen recovery plan. Compare R2 opens 31.00 and 31.20, then calculate the accepted path's net result at target 35.

INDEPENDENT PRACTICE

Audit the ambiguous account

Use P1's conventions on an unfamiliar fictional bar: a $20,000 account, 40 shares filled at $25.00, stop $24.00, target $27.00, $5 round-trip cost and stop-first resolution of an ambiguous bar. (a) The first held bar is O25.20/H27.30/L23.80/C26.10. Give 1R, the stop-first and target-first net results in dollars and R, the ending cash under each, and the size of the difference. (b) Suppose instead the bar is O23.90/H27.30/L23.80/C26.10. What does P1 record, and why is there no sequence ambiguity? (c) Write the review sentence that keeps the modeled result separate from what the bar can show.

INDEPENDENT PRACTICE

Transfer to sparse historical facts

The historical GME extract supplies closes and an intraday high but no full OHLC window or opening quote. Which pieces of P1 can be evaluated, and which facts would you request before modeling an order?

INDEPENDENT PRACTICE

Audit a calendar-selected historical cutoff

The gallery ledger lists SPY's April 2016 cell as adverse, but the chapter never works it. Using only the archive rows below (adjusted dollars, as supplied), (a) reproduce the prior-five mean and

the gate, (b) compute the unrounded upper and lower levels, (c) find the first resolving observation, and (d) explain what kind of resolution this is and what it does not establish about a historical stop execution.

SPY archive rows around the 2016-04-01 cutoff (CC0 archive, adjusted dollars)

Date	Open	High	Low	Close
2016-03-24	195.98	197.10	195.73	197.06
2016-03-28	197.54	197.78	196.68	197.18
2016-03-29	196.73	199.14	196.38	199.01
2016-03-30	200.15	200.72	199.46	199.88
2016-03-31	199.77	200.26	199.21	199.39
2016-04-01 (cutoff)	198.26	200.98	197.90	200.77
2016-04-04	200.68	200.91	199.75	200.10
2016-04-05	198.58	200.12	197.81	198.11

CHECK YOUR UNDERSTANDING

One question before you go

GAPS is cut from 30 to 15 shares with the same prices and a flat $5 round-trip fee. Which comparison is correct?

A. Both gross and net R remain −3.3333 because the share count halves.

B. Gross R remains −3.3333; net R changes from −3.3889 to −3.4444 because the fixed fee consumes more R.

C. Net R halves because dollar loss approximately halves.

Sources & further reading

- SEC / Investor.gov · Understanding Order Types ↗
  Market, limit, stop, and stop-limit mechanics; triggered stops do not guarantee execution price. Numerical plans and paths in the chapters are original fictional examples. Original access recorded 2026-10-01. Latest access check 2026-10-12: web reader accessible.

- SEC / Investor.gov · Investor Bulletin: Performance Claims ↗
  Scrutiny of hypothetical performance and performance presentations. Journal designs and fictional case calculations are original instructional examples. Original access recorded 2026-10-01. Latest access check 2026-10-12: web reader accessible.
- SEC staff (2021), Equity and Options Market Structure Conditions in Early 2021 ↗
  Dated GME observations in section 3.2; a retrospective staff report, not a contemporaneous signal feed. Original access recorded 2026-10-01. Latest access check 2026-10-12: web reader accessible.
- SEC · Webmaster Frequently Asked Questions, reuse of content ↗
  Government-created SEC content is free to access and reuse; some site content, such as stock art, is excluded. Not a license for underlying vendor data. Original access recorded 2026-10-01. Latest access check 2026-10-12: web reader accessible.
- Apple (April 23, 2014), Capital Return Program and Stock Split ↗
  Issuer announcement of a seven-for-one split and record/trading dates; no historical OHLC feed. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- Apple (January 26, 2016), Apple Reports Record First Quarter Results ↗
  Issuer declaration of a $0.52 per-share dividend payable February 11, 2016, to holders of record on February 8, 2016; the February 4 ex-date follows from the then two-business-day ex-date convention, not from this release. Original access recorded 2026-10-12. Latest access check 2026-10-12: retrieved.
- Boris Marjanovic / Kaggle · Huge Stock Market Dataset (version 3, CC0) ↗
  Depositor metadata explicitly applies CC0-1.0 and states dividend/split-adjusted daily OHLC. Local extracts preserve AAPL, MSFT and SPY 2016 rows. Upstream provider, exact session/time zone, adjustment algorithm and point-in-time revisions are not documented; no certified quote feed or execution/performance claim. Original access recorded 2026-10-02. Latest access check 2026-10-12: unavailable.
- Creative Commons · CC0 1.0 Universal deed and legal code ↗
  CC0 states the affirmer waives their copyright and related rights to the extent allowed. Other persons' rights and warranties are not supplied; this is the recorded permission basis, not chain-of-title certification. Original access recorded 2026-10-02. Latest access check 2026-10-12: retrieved.

Teaching examples are fictional unless labelled historical. Inspect this chapter's evidence. Sources support the specified concepts; they do not validate a profitable trading strategy.

From chart idea to trade plan ↗ Multiple timeframes: one market, several summaries ↗

Evidence before conviction ↗

Practice & review this part →